$$\int \frac{x+2}{\sqrt{2x+3}}dx = \int \frac{1}{2}\left\{(2x+3)^{\frac{1}{2}} + (2x+3)^{-\frac{1}{2}}\right\}dx$$

$$= \frac{1}{2}\left\{\frac{(2x+3)^{\frac{3}{2}}}{3} + (2x+3)^{\frac{1}{2}}\right\} + c$$

Example 2.8

Evaluate $\int \frac{1}{\sqrt{x+2}-\sqrt{x-2}}dx$

Solution:

By rationalisation,

$$\frac{1}{\sqrt{x+2}-\sqrt{x-2}} = \frac{1}{\sqrt{x+2}-\sqrt{x-2}} \times \frac{\sqrt{x+2}+\sqrt{x-2}}{\sqrt{x+2}+\sqrt{x-2}}$$

$$= \frac{\sqrt{x+2}+\sqrt{x-2}}{4}$$

$$\int \frac{1}{\sqrt{x+2}-\sqrt{x-2}}dx$$

$$= \int \frac{\sqrt{x+2}+\sqrt{x-2}}{4}dx$$

$$= \frac{1}{6}\left\{(x+2)^{\frac{3}{2}} + (x-2)^{\frac{3}{2}}\right\} + c$$

Exercise 2.1

Integrate the following with respect to x.

1. $\sqrt{3x+5}$
2. $\left(9x^2 - \frac{4}{x^2}\right)^2$
3. $(3+x)(2-5x)$
4. $\sqrt{x}\left(x^3 - 2x + 3\right)$
5. $\frac{8x+13}{\sqrt{4x+7}}$
6. $\frac{1}{\sqrt{x+1}+\sqrt{x-1}}$
7. If $f'(x) = x + b, f(1) = 5$ and $f(2) = 13$, then find $f(x)$
8. If $f'(x) = 8x^3 - 2x$ and $f(2) = 8$, then find $f(x)$

Type: II

(i) $\int \frac{1}{x}dx = \log|x| + c$

(ii) $\int \frac{1}{ax+b}dx = \frac{1}{a}\log|ax+b| + c$

Example 2.9

Evaluate $\int \frac{3x^2+2x+1}{x}dx$

Solution:

$$\int \frac{3x^2+2x+1}{x}dx = \int\left(3x+2+\frac{1}{x}\right)dx$$

$$= \frac{3x^2}{2} + 2x + \log|x| + c$$

Example 2.10

Evaluate $\int \frac{2}{3x+5}dx$

Solution:

$$\int \frac{2}{3x+5}dx = 2\int \frac{1}{3x+5}dx$$

$$= \frac{2}{3}\log|3x+5| + c$$

DO YOU KNOW?

$$\int \frac{a}{ax+b}dx = \int \frac{d(ax+b)}{ax+b}$$

$$= \log|ax+b| + c$$

Example 2.11

Evaluate $\int \frac{x^2+2x+3}{x+1}dx$

Solution:

Split into simple integrands

$$\frac{x^2+2x+3}{x+1} = \frac{(x^2+2x+1)+2}{x+1}$$

$$= (x+1) + \frac{2}{x+1}$$

$$\int \frac{x^2+2x+3}{x+1}dx = \int\left\{(x+1) + \frac{2}{x+1}\right\}dx$$

$$= \frac{x^2}{2} + x + 2\log|x+1| + c$$

Example 2.12

Evaluate $\int \frac{x^3+5x^2-9}{x+2}dx$

Solution:

By simple division,

$$\frac{x^3+5x^2-9}{x+2} = x^2 + 3x - 6 + \frac{3}{x+2}$$

XII Std - Business Maths & Stat EM Chapter 2.indd 30 2/28/2020 5:37:50 PM